Shreyas Narayanan

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Education

University of Michigan College of Engineering — Ann Arbor, MI
B.S.E. in Computer Engineering | GPA: 3.71 — *Expected December 2027*
- **Relevant Coursework**: Computer Architecture, Digital IC Design, Embedded Systems, Data Structures and Algorithms

Technical Skills

Programming: SystemVerilog, C/C++, Python, Assembly (ARM/RISC-V), Scripting (TCL, Perl), PyTorch
Tools: Cadence (JasperGold, Virtuoso), Synopsys (Verdi, VCS), Quartus Prime, ModelSim, Altium
Platforms: AXI, MESI/MOESI, MCUs (STM, Microchip), FPGAs (Altera), SoC, UVM, FreeRTOS, JTAG

Experience

Reditus Space | YC 25 — May 2026 – August 2026
Flight Software Intern
- Architected a NASA cFS propulsion stack with a Prop Manager routing 40 telemetry points and 20+ commands. Engineered a dedicated CANopen driver executing segmented SDO transfers against a controller node's Object Dictionary.
- Developed Raspberry Pi firmware for an Arducam IMX519 payload imaging pipeline with H.264 compression and developed F Prime GDS command/telemetry descriptors and a custom packetizer for remote real-time tuning during ground passes.

RISC-M — April 2026 – Present
Co-Founder and Technical Lead
- Co-founded UMich's first VLSI/Computer Architecture organization, directing team-wide RTL and Physical Design projects while connecting peers with industry leaders. Grew the club to over 40 members and 3 ongoing architecture projects.
- Designing an ML accelerator architecture featuring a 2D mesh NoC and an Ara-inspired lane-based vector unit using an RVV model to simulate Direct Compute Access logic that processes in-transit reduction packets within the network.
- Leading a team to implement the cycle-approximate simulator in Python and C++ while modeling the in-order scalar core and vector lane arbitration logic to benchmark collective operations against GPU-accelerated training and inference.

Interactive Sensing and Computing Lab — January 2025 – Present
Embedded Systems Engineering Intern
- Ported a low-power voice obfuscation algorithm from Python to C on an STM32G431KB, implementing Q31 fixed-point arithmetic and 6-band frequency recombination under tight RAM and I2S-to-DAC bit-width constraints.
- Validated obfuscation fidelity by reverse-engineering an Amazon Echo, soldering an analog injection line to the mic PCB, and driving it via DAC across 900 test conditions, preserving 88% of baseline Alexa recognition accuracy.
- Designed a real-time magnetic signature detection algorithm in C, leveraging digital signal processing to classify eyedropper cap movements across multiple states via a threshold-based state machine and achieving a 95% detection accuracy.
- Developed a fiber laser PCB prototyping process, achieving 0.25mm accuracy and cutting iteration time to 1.5 hours.

Introduction to Logic Design — August 2025 – December 2025
Instructional Aide
- Taught weekly labs for 30+ undergraduates, instructing on Verilog RTL modeling, FPGA synthesis, and digital logic.
- Mentored 20+ students via office hours and Piazza, guiding HDL debugging, RTL best practices, and core concepts.

Michigan Mars Rover Team — August 2024 – Present
Embedded Hardware Developer
- Designed Altium schematics and multi-layer PCBs for 18V/12V buck converters powering LDOs and CAN transceivers.
- Automated pre-production validation via SPICE parametric sweeps on PDB capacitor sizing to eliminate faults.

Projects

2-Way Superscalar Out-of-Order RISC Processor
- Co-designed a 2-way superscalar out-of-order RISC pipeline in SystemVerilog with a micro-op queue decoupling the fetch/dispatch stages from an execution backend featuring dual ALUs, a load/store unit, and a 4-wide register file.
- Integrated a banked BTB with a gshare predictor for speculative execution alongside L1 caches and a 2-entry victim cache, implementing micro-op decomposition of register-offset instructions to preserve strict memory access ordering.
- Swept cache and BTB sizes to map a Pareto optimal frontier, resolving critical path bottlenecks to hit a 11.9ns clock.

2-Axis Magnetic Sand Art Table
- Developed STM32L4R5 embedded firmware in C to control a 2-axis stepper motor gantry, executing real-time coordinate transformation algorithms to map analog joystick, touchscreen, and IR wand inputs to physical motor positions.
- Designed motion control system with PWM-driven stepper drivers to coordinate simultaneous dual-axis movement for smooth path execution, enabling real-time joystick control via ADC and touchscreen drawing via SPI interface.
- Implemented multi-mode operation with preset pattern generation, line segmentation algorithms to discretize touchscreen drawings into executable motor commands, and mode switching through FreeRTOS task priorities.